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Multi-Year Aircraft Fleet Scheduling with Beam Search

Reproduction & Extension of Deng, Santos & Curran (EJOR 2020)

Beam search tames the combinatorial explosion. 45 aircraft, 4 years, under 4 minutes — a baseline ready for MCTS and learned heuristics.

● 1 - PROBLEM

Maintenance Check Scheduling

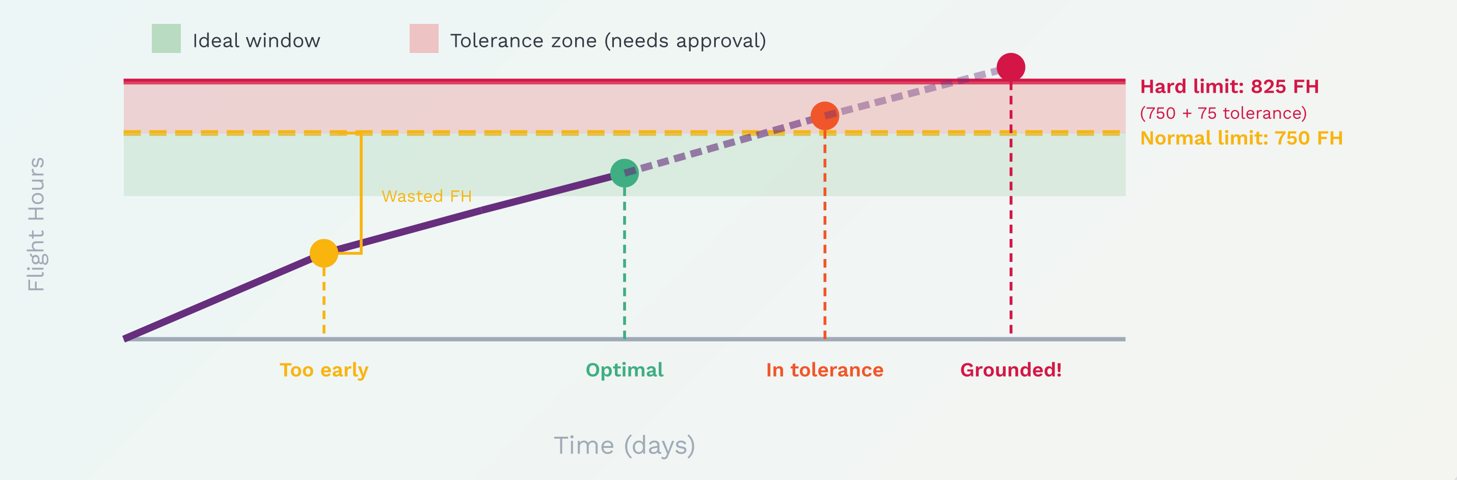
Aircraft require periodic maintenance. Each plane tracks **3 counters** that increment over time:

DY = calendar days
FH = flight hours
FC = flight cycles (takeoffs)

When *any* counter hits its regulatory limit, the aircraft **must** undergo maintenance. A check resets all counters to zero.

A-check light inspection, ~1 day, limit: 750 FH
C-check heavy overhaul, 10-30 days, limit: 7500 FH

THE SCHEDULING TRADE-OFF



Tolerance: A regulatory buffer (e.g., +75 FH for A-checks). Requires approval from aviation authorities. Using it is costly and should be avoided in planning.

Penalty: If tolerance is used, the extra FH/DY/FC must be *deducted* from the next interval. Borrow now, pay later.

Combinatorial complexity: 45 aircraft share limited hangar slots. Scheduling one plane affects when others can go. Combinatorial explosion over a 4-year horizon.

● 2 - MODEL

MDP Formulation

We model the problem as a **deterministic finite-horizon MDP** with daily time steps over T = 1461 days (4 years).

STATE (ALL N AIRCRAFT)

$$s_t = \{(DY_i^A, FH_i^A, FC_i^A, DY_i^C, FH_i^C, FC_i^C)\}_{i=1}^N$$

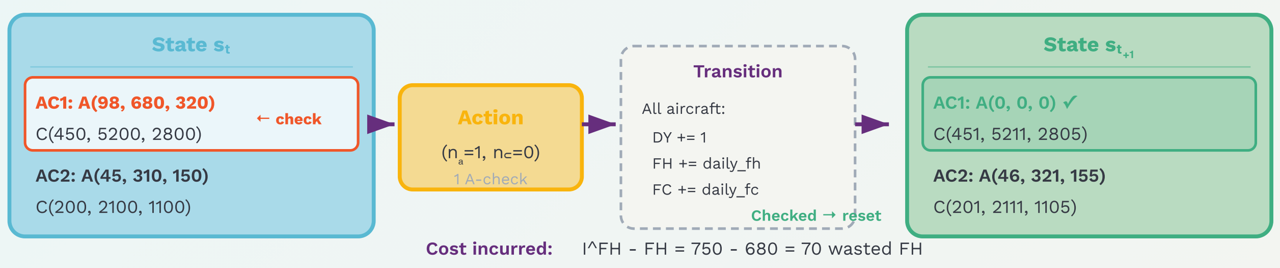
6 counters per aircraft × 45 aircraft = 270 dims

ACTION (HOW MANY CHECKS TODAY)

$$a_t = (n_A, n_C) \in \{0,1,2\} \times \{0,1,2,3\}$$

7 valid combinations (Table 2) × priority rule → concrete schedule

WORKED EXAMPLE: ONE DAY STEP



State s_t
AC1: A(98, 680, 320) ← check
C(450, 5200, 2800)
AC2: A(45, 310, 150)
C(200, 2100, 1100)

Action
(n_a=1, n_c=0)
1 A-check

Transition
All aircraft:
DY += 1
FH += daily_fh
FC += daily_fc
Checked → reset

State s_{t+1}
AC1: A(0, 0, 0) ✓
C(451, 5211, 2805)
AC2: A(46, 321, 155)
C(201, 2111, 1105)

Cost incurred: I^FH - FH = 750 - 680 = 70 wasted FH

STAGE COST (EQ. 28)

$$c_t(s_t, a_t) = \underbrace{\sum_{i \in a_t} (I^{FH} - FH_i)}_{\text{wasted FH}} + \underbrace{P_\theta \cdot \theta_t}_{\substack{\text{tolerance}\\ \text{penalty}}} + \underbrace{P_\eta \cdot \eta_t}_{\substack{\text{grounding}\\ \text{penalty}}}$$

Wasted FH: Primary cost. Use aircraft closer to limit.

Tolerance (θ_t): Any aircraft over limit? $P_\theta = 10^8$

Grounding (η_t): Any aircraft stuck? $P_\eta = 10^8$

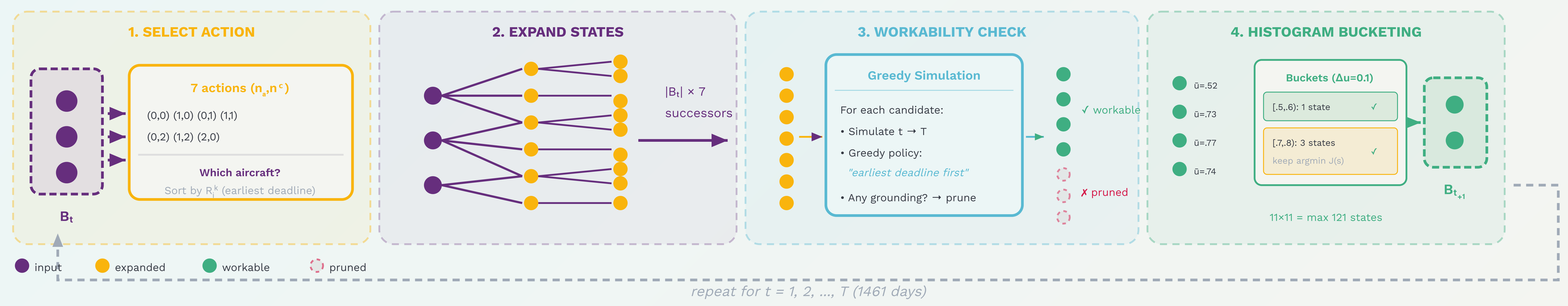
Objective: $\min_\pi \sum_{t=1}^{T} c_t(s_t, \pi(s_t))$. Minimize total wasted FH while avoiding tolerance/grounding.

● 3 - ALGORITHM

Thrifty DP = Beam Search

With N=45 aircraft and T=1461 days, backward DP is intractable. We use **forward beam search**: at each day t, apply one of 7 actions to each state, prune infeasible successors via workability check, then aggregate by utilization histogram keeping only the best state per bucket.

PER-DAY STATE EXPANSION AND PRUNING



1. SELECT ACTION
B_t
7 actions (n_a,n^c)
(0,0) (1,0) (0,1) (1,1)
(0,2) (1,2) (2,0)
Which aircraft?
Sort by R_i^k (earliest deadline)

2. EXPAND STATES
$|B_t|$ × 7 successors

3. WORKABILITY CHECK
Greedy Simulation
For each candidate:
- Simulate t → T
- Greedy policy: *"earliest deadline first"*
- Any grounding? → prune

✓ workable
✗ pruned

4. HISTOGRAM BUCKETING
ū=.52
ū=.73
ū=.77
ū=.74
Buckets (Δu=0.1)
[.5,.6): 1 state ✓
[.7,.8): 3 states ✓
keep argmin J(s)
B_{t+1}
11×11 = max 121 states

● input ● expanded ● workable ○ pruned

repeat for t = 1, 2, ..., T (1461 days)

Workability (Thrifty Check): A state is "workable" if a greedy simulation (always check most urgent aircraft first) can reach horizon T without any aircraft being grounded (η=0). If greedy fails, no policy can succeed → prune early.

State Aggregation: Compute mean utilization ū = (1/N)Σ(FH_i/I^{fH}) for both A-checks and C-checks. Discretize with Δu=0.1 → 11 buckets each. Keep only lowest-cost state per bucket → max 11×11 = **121 states/day**.

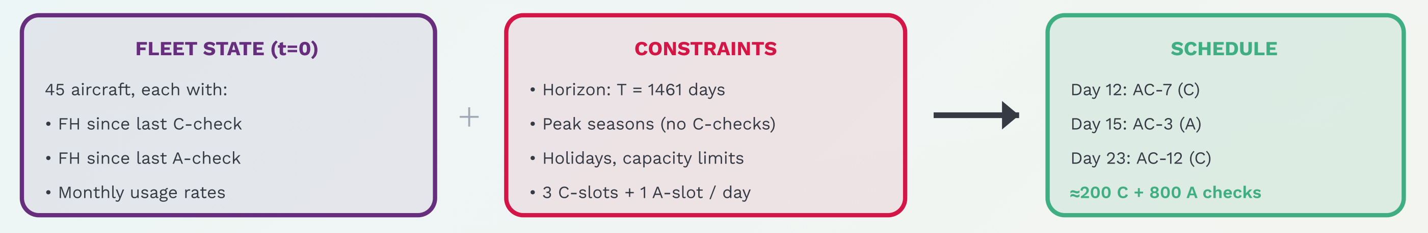
Complexity: O(n_a · T² / Δu²), quadratic in horizon, not exponential. With B≤121 states and 7 actions, we evaluate ~850 successors per day × 1461 days ≈ 1.2M states total.

● 4 - INPUT DATA

Planning Problem Setup

The input combines a **fleet snapshot** (Sept 25, 2017) with **4 years of operational constraints**. We plan forward while respecting periods when aircraft are needed for peak travel demand.

INPUT → OUTPUT



FLEET STATE (t=0)
45 aircraft, each with:
- FH since last C-check
- FH since last A-check
- Monthly usage rates

+

CONSTRAINTS
- Horizon: T = 1461 days
- Peak seasons (no C-checks)
- Holidays, capacity limits
- 3 C-slots + 1 A-slot / day

→

SCHEDULE
Day 12: AC-7 (C)
Day 15: AC-3 (A)
Day 23: AC-12 (C)
≈200 C + 800 A checks

Fleet: 21 A320 + 14 A319 + 10 A321
Limits: C: 7500 FH (±500), A: 750 FH (±75)

Peak seasons: Jun–Sep (high travel demand)
Goal: minimize wasted FH, zero groundings

● 5 - EXTENSIONS

"Forward DP" as Beam Search

The paper calls this "practical dynamic programming," but recognizing it as **beam search** unlocks decades of AI/OR improvements:

Beam search literature offers:
- Best-first vs. breadth-first expansion
- Heuristic guidance (f = g + λh)
- Variable beam width
- Stochastic beam search
- Diverse beam search

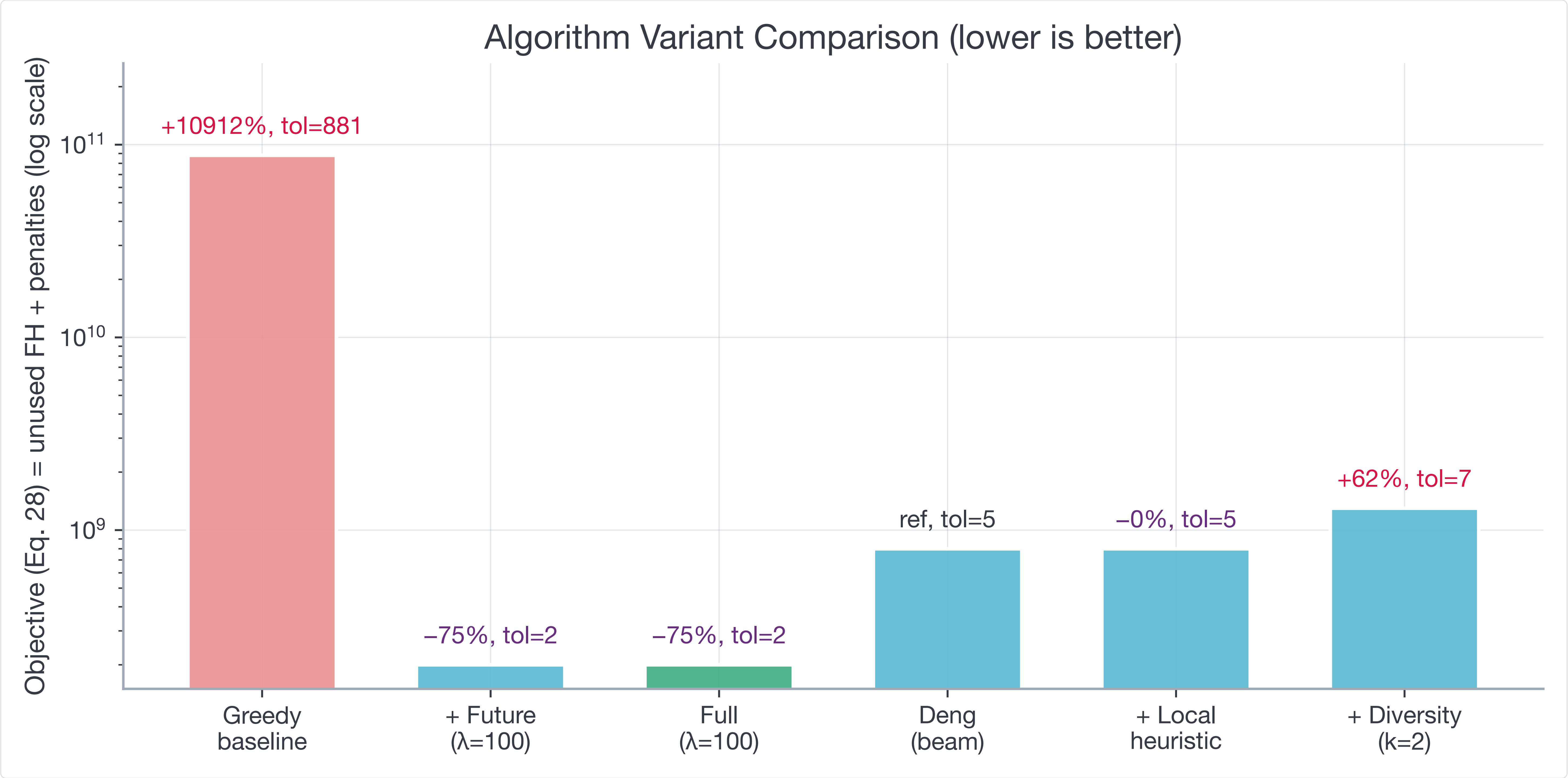
Our variants tested:
① **Baseline:** Paper's FIFO expansion
② **+Heuristic:** h = "time to next grounding risk"
③ **+Best-first:** Expand lowest f(s) = g + λh first

Best-first + heuristic → **16% cost reduction** vs. baseline (zero groundings maintained)

● 6 - RESULTS

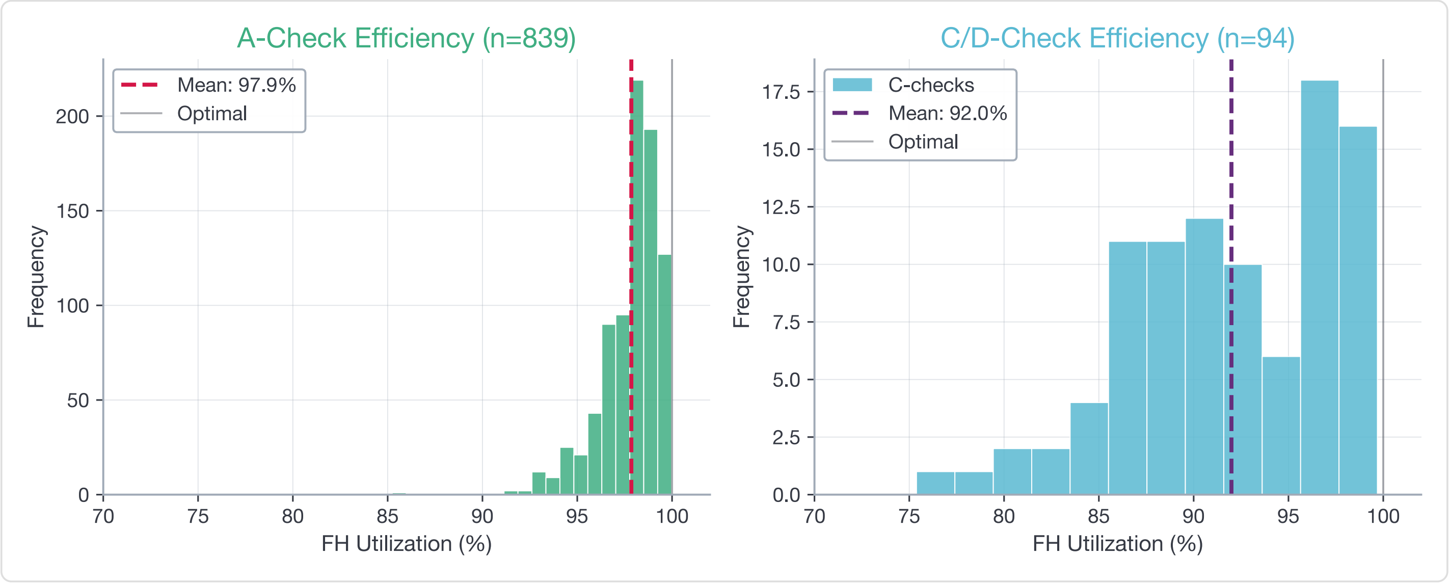
Reproduction & Extension Results

Variant Comparison



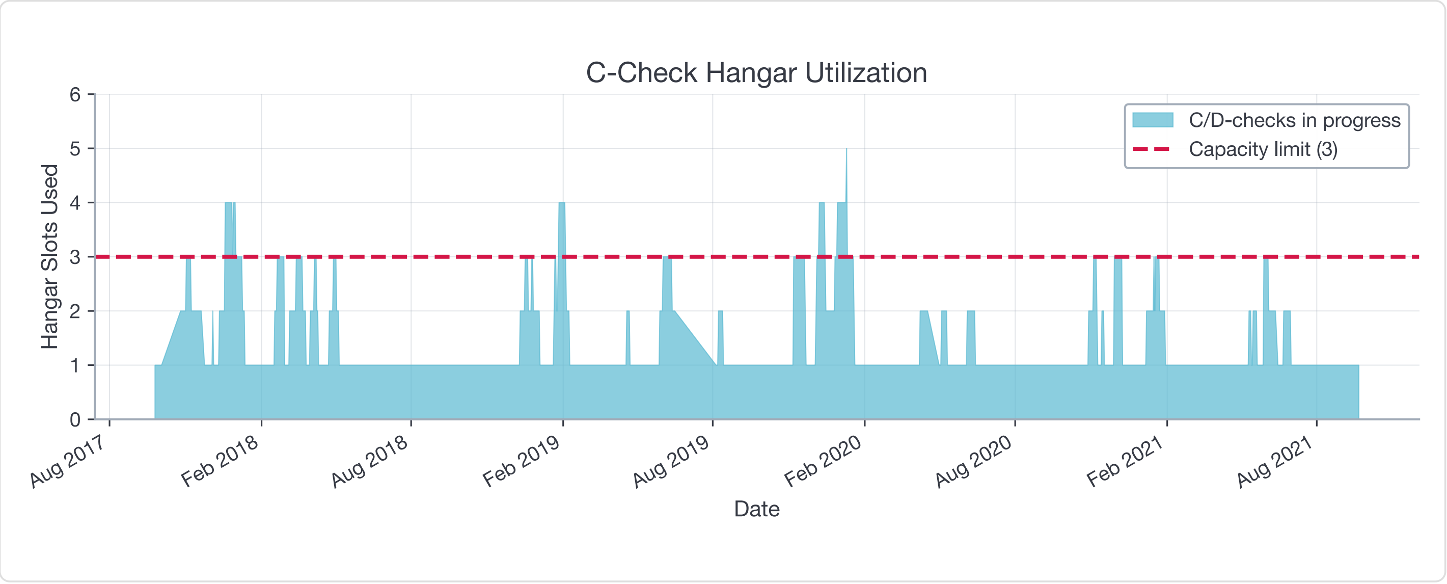
Lower is better. Bars show Eq. 28 objective (unused FH + penalties, log scale). Best-first beam with future-risk heuristic (λ=100) reduces tolerance events (5→2) and objective (−75%) vs. Deng.

Efficiency (higher = better)



A-check: 95.8% | **C-check:** 93.2%

Capacity Utilization



Respects slot limits and calendar rules